# Zhijing Wu

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## Education

**Sichuan University** Chengdu, China
*B.Eng. in Computer Science* *Sep. 2025 – Expected Jun. 2030*
- Overall GPA: **3.88/4.00**; Compulsory-course GPA: **4.00/4.00**; Compulsory-course average: **94.7/100**; Rank: **2/288**.

## Research Interests

LLM Agents and Structured Reasoning; Agent Memory and Planning; World Models and Generative Systems; Execution-Grounded Evaluation.

## Publications

Jinghong Tang, **Zhijing Wu**, Yikun Liu, Wei Ju, and Jiyao Wang. "Hyperbolic Self-Supervised Pretraining for Hierarchical EEG Representation Learning in Generalizable Parkinson's Disease Detection." *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026).* **Regular Paper, Accepted**.

## Selected Research Projects

**Guided Topology Diffusion: Execution-Semantics Audit** | *Independent Study* 2026
- Traced generated communication graphs through runtime construction and execution in the released GTD/GSM8K pipeline. Across **603 records**, coordinate canonicalization increased mean raw-to-executed edge Jaccard from **0.152 to 0.863**; matched downstream task correctness remained 0.96 vs. 0.96.

**Cardiff: Controlled Diffusion Sampling and Geometry** | *Independent Study* Oct. 2026
- Compared released Porto GPS-stage diffusion sampling intervals 20/50 on **64 conditions** × **3 seeds** with fixed Stage 1 latents; observed **2.503× median paired-batch GPS-stage speedup** (conditional 95% interval 2.432–2.692×). Length/step summaries met frozen point-estimate margins, while turning tails varied; road validity was not evaluated.

**AutumnBench: Execution and Scoring Semantics** | *Independent Audit* Oct. 2026
- Verified **117/117** released visible frames in a CPU-only MFP replay and traced historical frame selection through Change Detection scoring in pinned public code; identified a source-grounded interface/scoring discrepancy. Not a full benchmark reproduction or a validated scorer repair.

**AgentHijack: Within-Step Corruption Timing** | *Independent Extension* 2026
- Compared perturbation boundaries over **3 Chrome tasks / 36 runs**; post-observation/pre-execution timing increased mean episode lengths by 4.00, 1.75 and 1.00 steps. Small task set and uncontrolled seeds limit inference.

**Language Modeling Foundations** | *Stanford CS336 Independent Study* 2026
- Implemented BPE tokenizer, Transformer components and training utilities in PyTorch using public course materials.

## Awards & Honors

**First Prize (Sichuan Division)**, National College Student Mathematics Competition, 2026.

**Meritorious Winner (M Award)**, Mathematical Contest in Modeling (MCM), May 2026.

## Technical Skills

**Programming / ML:** Python, C/C++, PyTorch, PyTorch Geometric, Transformers
**Research:** LLM Agents, Multi-Agent Systems, Agent Memory, Controlled Evaluation, Generative Modeling
**Systems / Tools:** Linux, Git, Docker, Conda, Reproducible Experimentation